

Citations of Retracted ICPSR Papers

Number of citations:

```
# add file id column to merged - TODO: move to dc/preprocessing file?
merged <- merged |> mutate(file.id = NA, .before = Title)

merged$file.id[which(merged$Title ==
  "Worlds Apart? The Reception of Genetically Modified Foods in Europe and the U.S."
)] = "WorldsApart"
merged$file.id[which(merged$Title ==
  "Nonmetropolitan sex-role ideologies: A longitudinal study"
)] = "NonmetroRole"
merged$file.id[which(merged$Title ==
  "School social bonds, school climate, and school misbehavior: A multilevel analysis"
)] = "SchoolSocialBonds"
merged$file.id[which(merged$Title ==
  "Suicide after natural disasters"
)] = "SuicideNatDisasters"
merged$file.id[which(merged$Title ==
  "Electronic cigarette use and myocardial infarction among adults in the US Population Assessment of T
)] = "ECigUse"
merged$file.id[which(merged$Title ==
  "The verdict is in: How did they decide? Using drivers' self-reported data to understand officers' de
)] = "HowDidTheyDecide"
merged$file.id[which(merged$Title ==
  "Chronic adolescent marijuana use as a risk factor for physical and mental health problems in young a
)] = "MarijuanaUse"
merged$file.id[which(merged$Title ==
  "Tuning in, not turning out: Evaluating the impact of ethnic television on political participation"
)] = "TurningIn"
merged$file.id[which(merged$Title ==
  "Implications of new hypertension guidelines in the United States"
)] = "HypertensionGuidelines"

# add retraction dates to refs
refs <- refs |> mutate(ret.date = NA, .after = year)

for (r in 1:nrow(refs)) {
  ix = which(refs$filepath[r] == merged$file.id)
  refs$ret.date[r] = merged$RetractionDate[ix]
}

# add retraction years to refs
refs <- refs |> mutate(ret.year = NA, .after = ret.date)

for (r in 1:nrow(refs)) {
  ix = which(refs$filepath[r] == merged$file.id)
  refs$ret.year[r] = merged$RetractionYear[ix]
```

```
}

# number of citations per paper
refs |> group_by(filepath) |> count()
```

```
## # A tibble: 9 x 2
## # Groups:   filepath [9]
##   filepath          n
##   <chr>          <int>
## 1 ECigUse         61
## 2 HowDidTheyDecide 4
## 3 HypertensionGuidelines 50
## 4 MarijuanaUse     80
## 5 NonmetroRole     11
## 6 SchoolSocialBonds 267
## 7 SuicideNatDisasters 178
## 8 TurningIn        16
## 9 WorldsApart      465
```

```
# number of citations per paper by year
refsbyyear <- refs |> group_by(filepath, year, ret.year) |>
  summarise(num_citations = n())
```

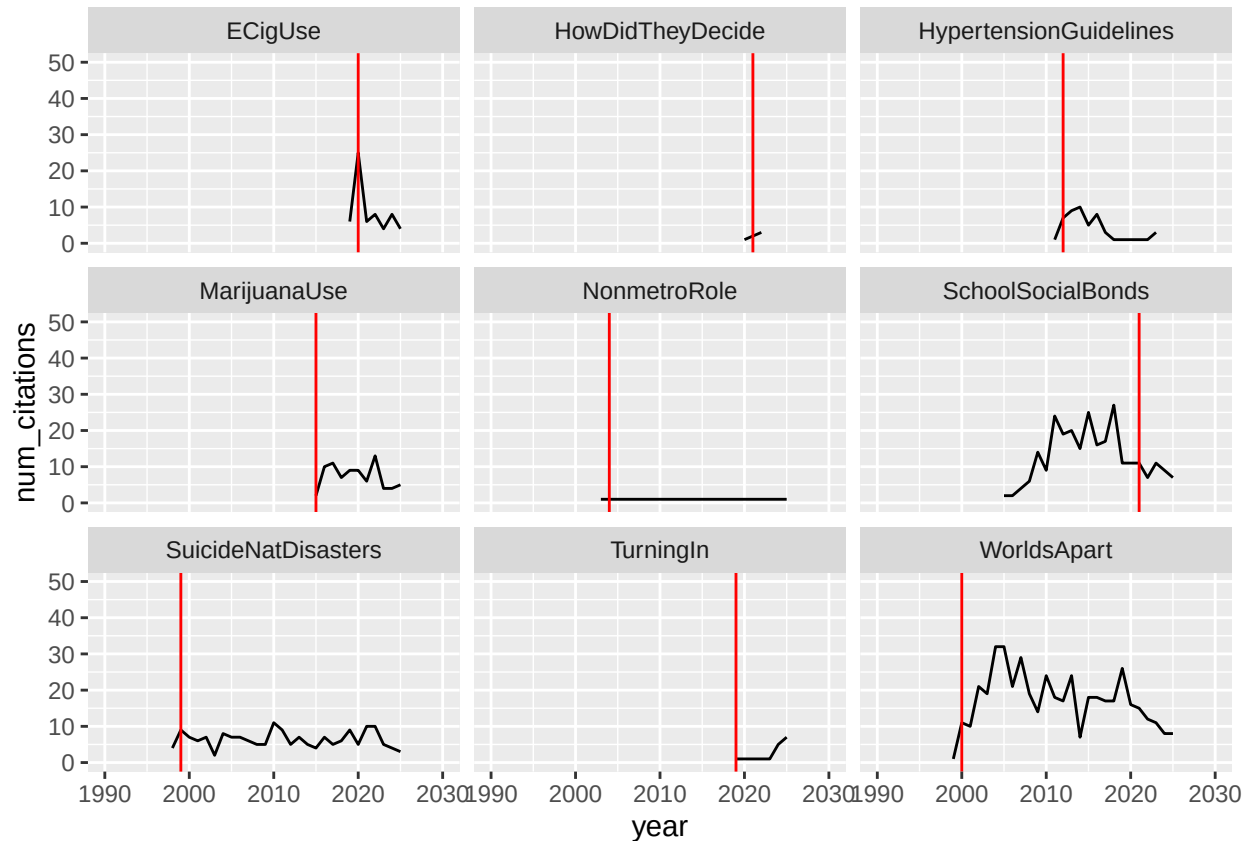
```
## `summarise()` has grouped output by 'filepath', 'year'. You can override using
## the `groups` argument.
```

```
(head(refsbyyear, 20))
```

```
## # A tibble: 20 x 4
## # Groups:   filepath, year [20]
##   filepath          year ret.year num_citations
##   <chr>          <dbl>   <dbl>         <int>
## 1 ECigUse         2019     2020             6
## 2 ECigUse         2020     2020            25
## 3 ECigUse         2021     2020             6
## 4 ECigUse         2022     2020             8
## 5 ECigUse         2023     2020             4
## 6 ECigUse         2024     2020             8
## 7 ECigUse         2025     2020             4
## 8 HowDidTheyDecide 2020     2021             1
## 9 HowDidTheyDecide 2022     2021             3
## 10 HypertensionGuidelines 2011     2012             1
## 11 HypertensionGuidelines 2012     2012             7
## 12 HypertensionGuidelines 2013     2012             9
## 13 HypertensionGuidelines 2014     2012            10
## 14 HypertensionGuidelines 2015     2012             5
## 15 HypertensionGuidelines 2016     2012             8
## 16 HypertensionGuidelines 2017     2012             3
## 17 HypertensionGuidelines 2018     2012             1
## 18 HypertensionGuidelines 2019     2012             1
## 19 HypertensionGuidelines 2021     2012             1
## 20 HypertensionGuidelines 2022     2012             1
```

Plots of citations by year. Retraction Year in red.

```
# plot citations by year
refsbyyear |>
  group_by(filepath) |>
  ggplot(aes(x = year, y = num_citations)) +
    geom_line() +
    geom_vline(aes(xintercept = ret.year), colour = "red") +
    xlim(1990, 2030) +
    ylim(0, 50) +
    facet_wrap(~filepath)
```



Summary table - number of citations before and after retraction year:

```
refsbyyear |> group_by(filepath, year <= ret.year) |>
  count()
```

```
## # A tibble: 18 x 3
## # Groups:   filepath, year <= ret.year [18]
##   filepath      `year <= ret.year`     n
##   <chr>          <lgl>             <int>
## 1 ECigUse        FALSE                5
## 2 ECigUse        TRUE                 2
## 3 HowDidTheyDecide FALSE                1
## 4 HowDidTheyDecide TRUE                 1
## 5 HypertensionGuidelines FALSE            10
## 6 HypertensionGuidelines TRUE                 2
## 7 MarijuanaUse   FALSE            10
## 8 MarijuanaUse   TRUE                 1
## 9 NonmetroRole   FALSE                 9
```

```
## 10 NonmetroRole      TRUE      2
## 11 SchoolSocialBonds FALSE      4
## 12 SchoolSocialBonds TRUE     17
## 13 SuicideNatDisasters FALSE    26
## 14 SuicideNatDisasters TRUE      2
## 15 TurningIn         FALSE      5
## 16 TurningIn         TRUE        1
## 17 WorldsApart       FALSE     25
## 18 WorldsApart       TRUE         2
```

Flag whether a retracted article was cited the same year it was retracted:

```
# create flag
refs <- refs |> mutate(ref.same.year = (year == ret.year))

# number citing papers retracted during the publication year
refs |> group_by(ref.same.year) |> summarise(n())
```

```
## # A tibble: 2 x 2
##   ref.same.year `n()`
##   <lgl>         <int>
## 1 FALSE         1065
## 2 TRUE           67
```

```
# number with available publication date
refs |> group_by(ref.same.year, !is.na(pub.date)) |> summarise(n())
```

`summarise()` has grouped output by 'ref.same.year'. You can override using the
`.groups` argument.

```
## # A tibble: 4 x 3
## # Groups:   ref.same.year [2]
##   ref.same.year `!is.na(pub.date)` `n()`
##   <lgl>         <lgl>         <int>
## 1 FALSE         FALSE         810
## 2 FALSE         TRUE          255
## 3 TRUE          FALSE          35
## 4 TRUE          TRUE           32
```

Pulling citation data manually for retractions for which `ret.same.year == TRUE`:

```
# setwd(...)

# refs |> filter(ref.same.year == TRUE) |>
#   select(title, authors.list, year, pub.date, doi) |>
#   write.csv(file = "refs_same_year.csv")

# manually pull and add dates to sheet, export to CSV

# Searched on WOS and Scopus. Could typically find data on WOS
# directly, on Scopus, viewed article page on Publisher
# website or on Proquest/SagePub/EBSCOHost through UM library
```

Manually downloaded data:

```
# read edited file into new df
refs_manual <- read.csv("refs_same_year_final.csv")
```

Reformat pub.date

```

month.names <- c(
  'JAN' = 1,
  'FEB' = 2,
  'MAR' = 3,
  'APR' = 4,
  'MAY' = 5,
  'JUN' = 6,
  'JUL' = 7,
  'AUG' = 8,
  'SEP' = 9,
  'OCT' = 10,
  'NOV' = 11,
  'DEC' = 12
)

get.date <- function(datestr) {
  # split into vector of (MONTH, [DAY], YEAR)
  datevec = str_split(datestr, pattern = '\\s') |> unlist()

  # convert year, month to numeric
  # check for date, if yes, convert to numeric
  year = datevec[length(datevec)]
  month = month.names[datevec[1]] |> unname()
  day = ifelse(length(datevec) > 2, datevec[2], NA)

  return(c(year, month, day))
}

refs$pub.year = numeric(nrow(refs))
refs$pub.month = numeric(nrow(refs))
refs$pub.day = numeric(nrow(refs))

for (r in 1:nrow(refs)) {
  # if pub.date data is provided, use get.date function
  if (!is.na(refs$pub.date[r])) {
    datevec = get.date(refs$pub.date[r])
    refs$pub.year[r] = ifelse(refs$year[r] == datevec[1],
                              refs$year[r],
                              datevec[1])
    refs$pub.month[r] = datevec[2]
    refs$pub.day[r] = datevec[3]
  }
  # if pub.date is not provided, use manually pulled data if available
  else {
    # check if doi is available
    if (!is.na(refs$doi[r])) {
      ix = which(refs$doi[r] == refs_manual$doi)
    }
    # else compare titles, publication years for match
    else {
      ix = which((refs$title[r] == refs_manual$title) &
                 (refs$year[r] == refs_manual$year))
    }
  }
}

```

```

# check whether replacement exists
if (length(ix) == 0) {
  # if not, re-use refs$year, insert NAs for month/day
  refs$pub.year[r] = refs$year[r]
  refs$pub.month[r] = NA
  refs$pub.day[r] = NA
}
# if so, replace
else {
  refs$pub.year[r] = refs_manual$pub.year.manual[ix]
  refs$pub.month[r] = refs_manual$pub.month.manual[ix]
  refs$pub.day[r] = refs_manual$pub.date.manual[ix]
}
}
}

```

Flag if month AND year of ref publication are same as retraction month and year

```

# add retraction months to refs
refs <- refs |> mutate(
  ret.month = mdy_hm(ret.date) %>% month(),
  .after = ret.year
)

# add retraction date specifically to refs
refs <- refs |> mutate(
  ret.day = mdy_hm(ret.date) %>% mday(),
  .after = ret.month
)

# number citing papers retracted during the publication month
refs |> group_by((year == ret.year) & (pub.month == ret.month)) |> summarise(n())

```

```

## # A tibble: 3 x 2
##   `(year == ret.year) & (pub.month == ret.month)` `n()`
##   <lgl>                                     <int>
## 1 FALSE                                     1122
## 2 TRUE                                      7
## 3 NA                                       3

```

More granular before/after retraction counts - only 3 refs published in retraction year for which no month could be obtained (1 publication citing “Suicide after natural disasters...”, 2 citing “Worlds Apart? ...”)

```

refs |> group_by(filepath, ((year < ret.year) | ((year == ret.year) & (pub.month < ret.month)))) |>
  count()

```

```

## # A tibble: 20 x 3
## # Groups:   filepath, ((year < ret.year) | ((year == ret.year) & (pub.month <
## #   ret.month))) [20]
##   filepath                                     ((year < ret.year) | ((year == ret.year) & (pub.month <
##   <chr>                                     <lgl>                                     <int>
## 1 ECigUse                                     FALSE                                     47
## 2 ECigUse                                     TRUE                                      14
## 3 HowDidTheyDecide                           FALSE                                      3
## 4 HowDidTheyDecide                           TRUE                                      1

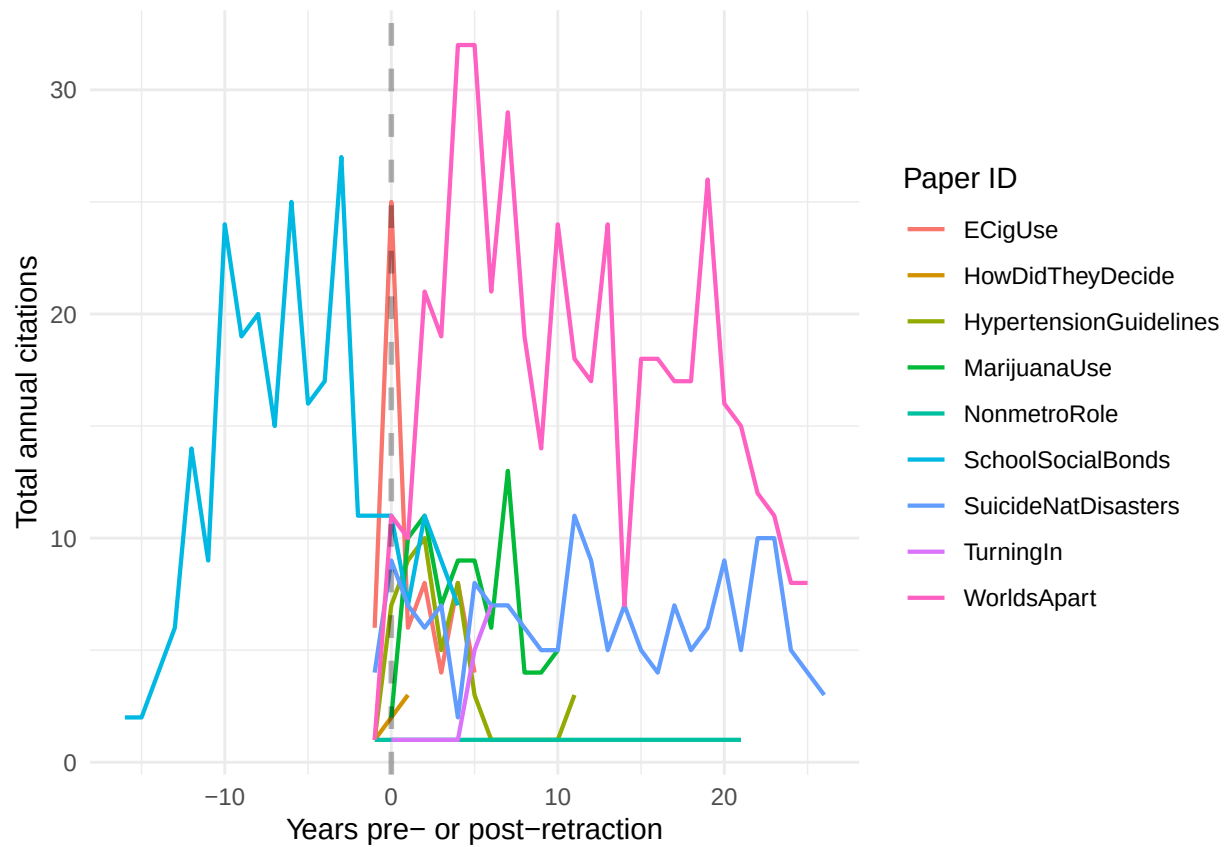
```

```
## 5 HypertensionGuidelines FALSE 44
## 6 HypertensionGuidelines TRUE 6
## 7 MarijuanaUse FALSE 79
## 8 MarijuanaUse TRUE 1
## 9 NonmetroRole FALSE 10
## 10 NonmetroRole TRUE 1
## 11 SchoolSocialBonds FALSE 45
## 12 SchoolSocialBonds TRUE 222
## 13 SuicideNatDisasters FALSE 173
## 14 SuicideNatDisasters TRUE 4
## 15 SuicideNatDisasters NA 1
## 16 TurningIn FALSE 15
## 17 TurningIn TRUE 1
## 18 WorldsApart FALSE 456
## 19 WorldsApart TRUE 7
## 20 WorldsApart NA 2
## # i abbreviated name:
## # 1: `((year < ret.year) | ((year == ret.year) & (pub.month < ret.month)))`
```

Normalize retraction year, line graph citations against normalized retraction year:

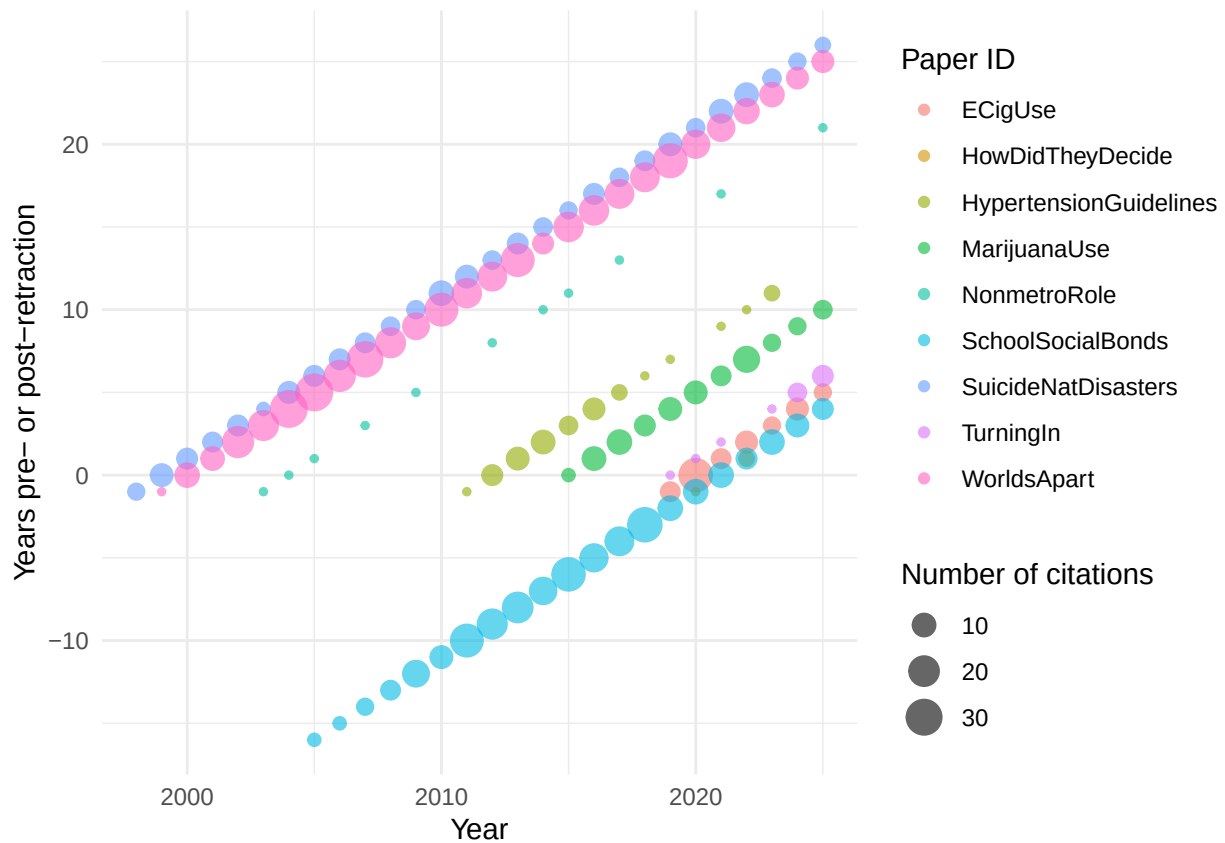
```
refs <- refs |> mutate(pub.year.norm = year - ret.year)

refs |> group_by(filepath, pub.year.norm) |>
  count() |>
  ggplot(aes(x = pub.year.norm, y = n, color = filepath)) +
    geom_line(linewidth = 0.75) +
    geom_vline(aes(xintercept = 0), linewidth = 1, linetype = "dashed",
               color = "black", alpha = 0.3) +
    labs(x = "Years pre- or post-retraction", y = "Total annual citations",
         color = "Paper ID") +
    theme_minimal()
```



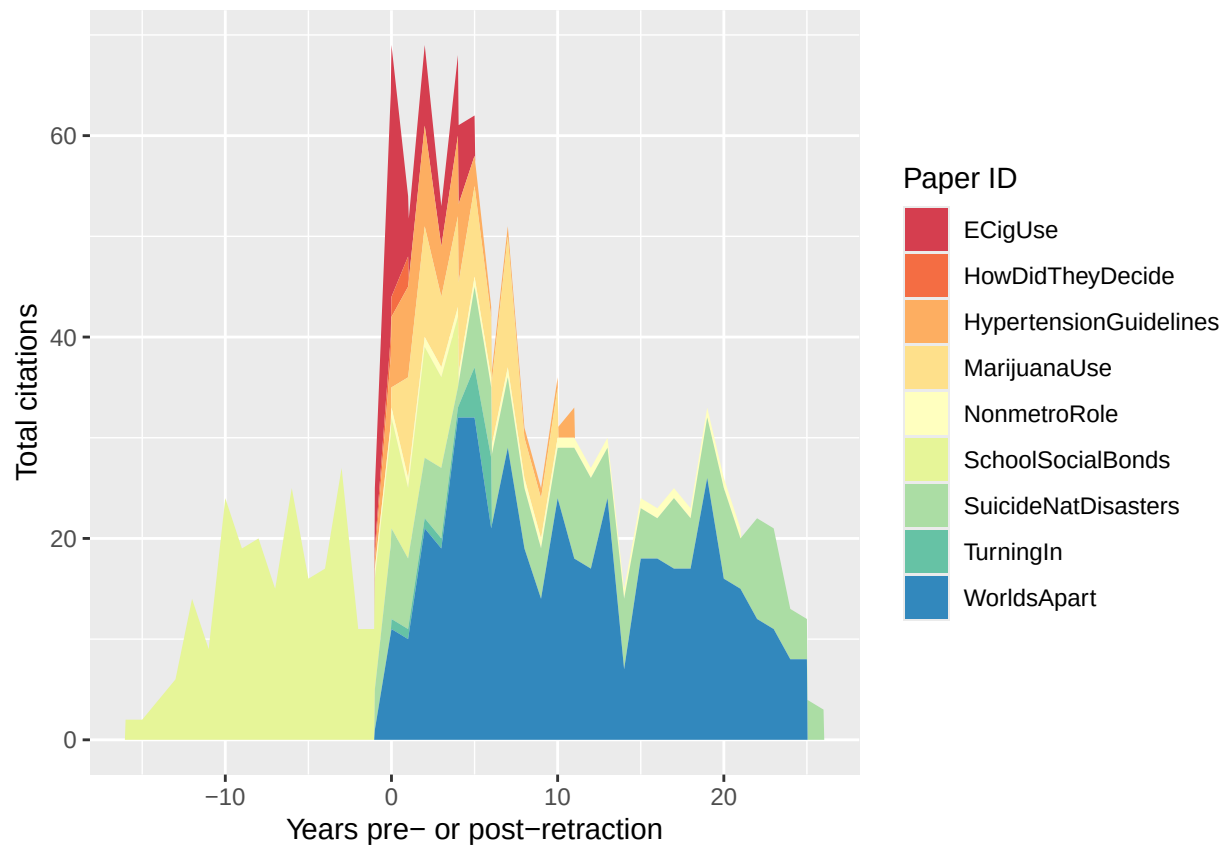
Bubble plot:

```
refs |> group_by(filepath, year, pub.year.norm) |>
  count() |>
  ggplot(aes(x = year, y = pub.year.norm, size = n, colour = filepath)) +
    geom_point(alpha = 0.6) +
    labs(x = "Year", y = "Years pre- or post-retraction",
         size = "Number of citations", colour = "Paper ID") +
    theme_minimal()
```

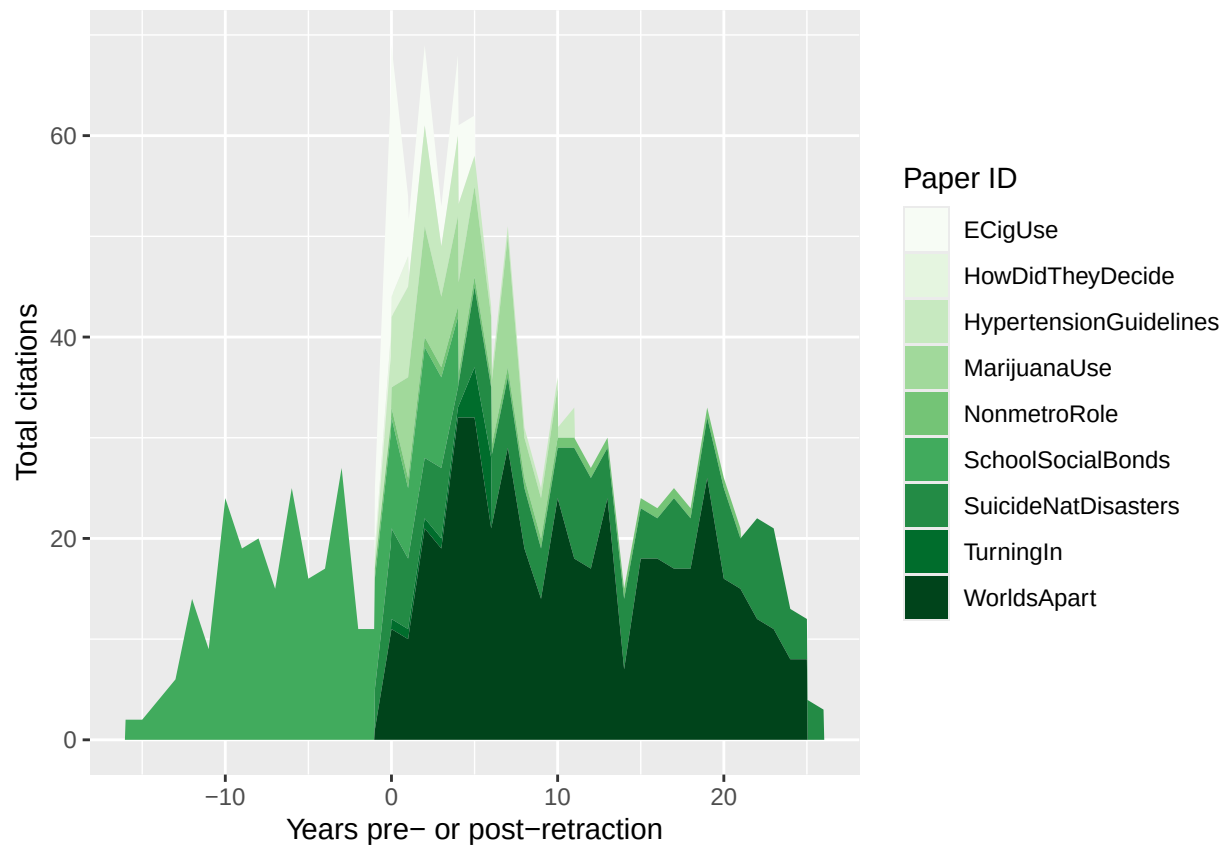


Area chart:

```
# color palette: spectral
refs |> group_by(filepath, year, pub.year.norm) |>
  count() |>
  ggplot(aes(x = pub.year.norm, y = n, fill = filepath)) +
  geom_area() +
  scale_fill_brewer(palette = "Spectral") +
  labs(x = "Years pre- or post-retraction",
       y = "Total citations",
       fill = "Paper ID")
```



```
# color palette: Purple/green
refs |> group_by(filepath, year, pub.year.norm) |>
  count() |>
  ggplot(aes(x = pub.year.norm, y = n, fill = filepath)) +
  geom_area() +
  scale_fill_brewer(palette = "Greens") +
  labs(x = "Years pre- or post-retraction",
       y = "Total citations",
       fill = "Paper ID")
```



Monthly citation rates: